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In [14]: import pandas as pd
df=pd.read_csv('breast_cancer_clinical3.csv')
print('breast_cancer_clinical3读取成功')
print('-----')
print('Univariate Logistic Regression Analysis:')
import statsmodels
import statsmodels.api as sm
import numpy as np
variables=[
    'Age',
    'Tumor_Size',
    'Ki67',
    'ER',
    'PR',
    'HER2'
]
results=[]
for variable in variables:
    y=df['Lymph_Node']
    X=df[[variable]]
    X=sm.add_constant(X)
    model=sm.Logit(y,X).fit(dis= False)
    beta=model.params[variable]
    p=model.pvalues[variable]
    OR=np.exp(model.params[variable])
    CI=np.exp(model.conf_int().loc[variable])
    results.append([
        variable,
        beta,
        OR,
        CI[0],
        CI[1],
        p
    ])
table1=pd.DataFrame(
    results,
    columns=[
        'Variable',
        'Beta',
        'OR',
        '95%CI上限',
        '95%CI下限',
        'p'
    ])
print(table1)
table1.to_excel('Univariate Logistic Regression Analysis.xlsx')
print('Univariate Logistic Regression Analysis.xlsx保存成功')
print('-----')
print('multivariate Logistic Regression Analysis:')
y=df['Lymph_Node']
X=df[['Age','Tumor_Size','Ki67','ER','PR','HER2']]
X=sm.add_constant(X)
model=sm.Logit(y,X).fit(dis= False)
print('Beta:')
beta=model.params
print(beta)
p=model.pvalues
print('P')

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print(p)
OR=np.exp(model.params)
print('OR:')
print(OR)
CI=np.exp(model.conf_int())
print(CI[0])
print('CI[1]:')
print(CI[1])
table2=pd.DataFrame({
    'Beta':beta,
    'OR':OR,
    '95%CI上限':CI[0],
    '95%CI下限':CI[1],
    'P':p
})
print('方式一table2:')
print(table2)
print('单因素Logistic回归分析显示，年龄、肿瘤大小、PR、HER2均与淋巴结转移有统计学
table2.to_excel('multivariate Logistic Regression Analysis.xlsx')
print('multivariate Logistic Regression Analysis.xlsx保存成功')
results1=[]
for variable in variables:
    beta=model.params[variable]
    OR=np.exp(model.params[variable])
    CI=np.exp(model.conf_int().loc[variable])
    p=model.pvalues[variable]
    results1.append([
        variable,
        beta,
        OR,
        CI[0],
        CI[1],
        p
    ])
table3=pd.DataFrame(
    results1,
    columns=[
        'Variable',
        'Beta',
        'OR',
        '95%CI上限',
        '95%CI下限',
        'P'
    ])
print('方式二table3:')
print(table3)
print('多因素Logistic回归分析结果分析显示，P值都大于0.05，因此都不是独立相关因素'

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breast_cancer_clinical3读取成功

Univariate Logistic Regression Analysis:

	Variable	Beta	OR	95%CI上限	95%CI下限	P
0	Age	0.284580	1.329203	1.036229	1.705010	0.025084
1	Tumor_Size	2.604026	13.518047	1.534596	119.078654	0.018989
2	Ki67	0.010215	1.010267	0.980639	1.040791	0.501194
3	ER	-2.261763	0.104167	0.009509	1.141126	0.064044
4	PR	-2.756840	0.063492	0.007072	0.570034	0.013821
5	HER2	2.233592	9.333333	1.193455	72.990686	0.033296

Univariate Logistic Regression Analysis.xlsx保存成功

multivariate Logistic Regression Analysis:

Beta:

const	-9.627728
Age	-0.819505
Tumor_Size	14.231241
Ki67	0.060029
ER	10.492835
PR	-3.830561
HER2	-0.178382

dtype: float64

P

const	0.764529
Age	0.497150
Tumor_Size	0.241103
Ki67	0.184782
ER	0.308264
PR	0.429511
HER2	0.980101

dtype: float64

OR:

const	6.587655e-05
Age	4.406498e-01
Tumor_Size	1.515478e+06
Ki67	1.061867e+00
ER	3.605624e+04
PR	2.169745e-02
HER2	8.366226e-01

dtype: float64

const	2.881299e-32
Age	4.137368e-02
Tumor_Size	7.026552e-05
Ki67	9.717194e-01
ER	6.178724e-05
PR	1.618976e-06
HER2	6.838726e-07

Name: 0, dtype: float64

CI[1]:

const	1.506168e+23
Age	4.693135e+00
Tumor_Size	3.268563e+16
Ki67	1.160378e+00
ER	2.104079e+13
PR	2.907884e+02
HER2	1.023491e+06

Name: 1, dtype: float64

方式一table2:

	Beta	OR	95%CI上限	95%CI下限	P
const	-9.627728	6.587655e-05	2.881299e-32	1.506168e+23	0.764529

Age	-0.819505	4.406498e-01	4.137368e-02	4.693135e+00	0.497150
Tumor_Size	14.231241	1.515478e+06	7.026552e-05	3.268563e+16	0.241103
Ki67	0.060029	1.061867e+00	9.717194e-01	1.160378e+00	0.184782
ER	10.492835	3.605624e+04	6.178724e-05	2.104079e+13	0.308264
PR	-3.830561	2.169745e-02	1.618976e-06	2.907884e+02	0.429511
HER2	-0.178382	8.366226e-01	6.838726e-07	1.023491e+06	0.980101

单因素Logistic回归分析显示，年龄、肿瘤大小、PR、HER2均与淋巴结转移有统计学关联。
multivariate Logistic Regression Analysis.xlsx保存成功

方式二table3:

	Variable	Beta	OR	95%CI上限	95%CI下限	P
0	Age	-0.819505	4.406498e-01	4.137368e-02	4.693135e+00	0.497150
1	Tumor_Size	14.231241	1.515478e+06	7.026552e-05	3.268563e+16	0.241103
2	Ki67	0.060029	1.061867e+00	9.717194e-01	1.160378e+00	0.184782
3	ER	10.492835	3.605624e+04	6.178724e-05	2.104079e+13	0.308264
4	PR	-3.830561	2.169745e-02	1.618976e-06	2.907884e+02	0.429511
5	HER2	-0.178382	8.366226e-01	6.838726e-07	1.023491e+06	0.980101

多因素Logistic回归分析结果分析显示，P值都大于0.05，因此都不是独立相关因素

In []: